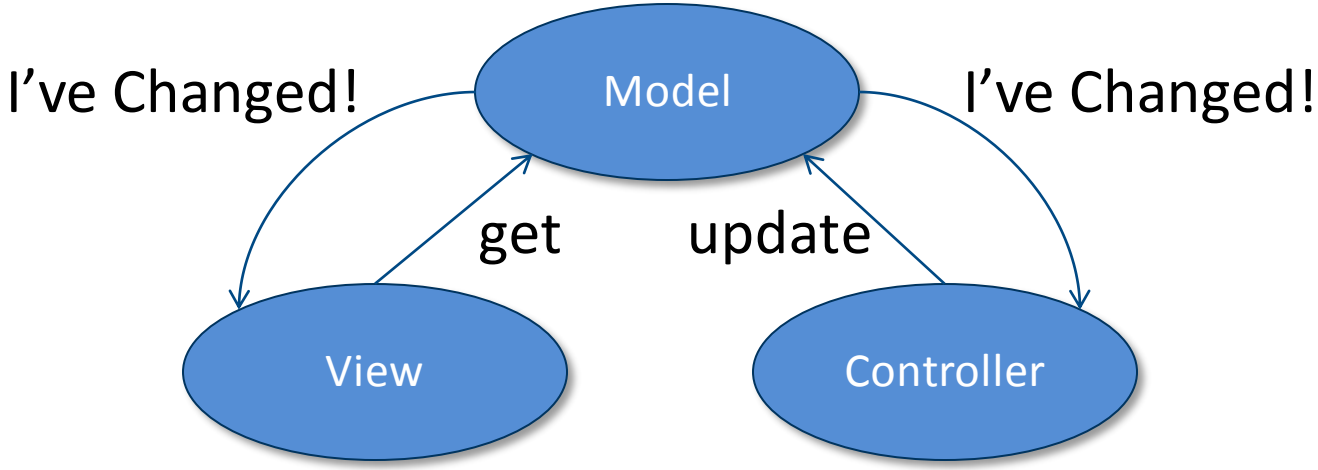
605.486 – Android Development

Module 4a
Lists

Model / View / Controller

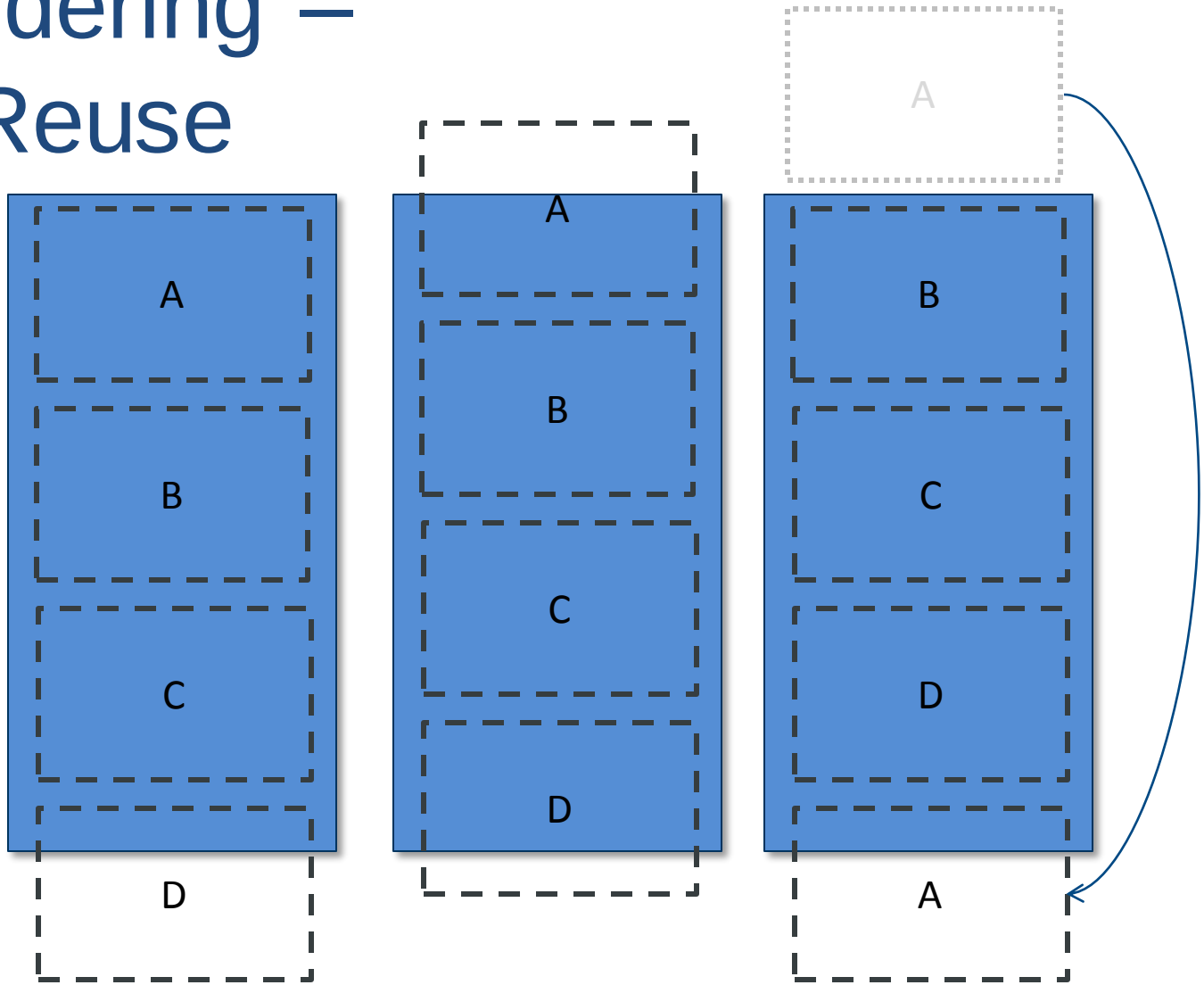


- ListAdapter is the model (+ row rendering)
- ListView is View/Controller

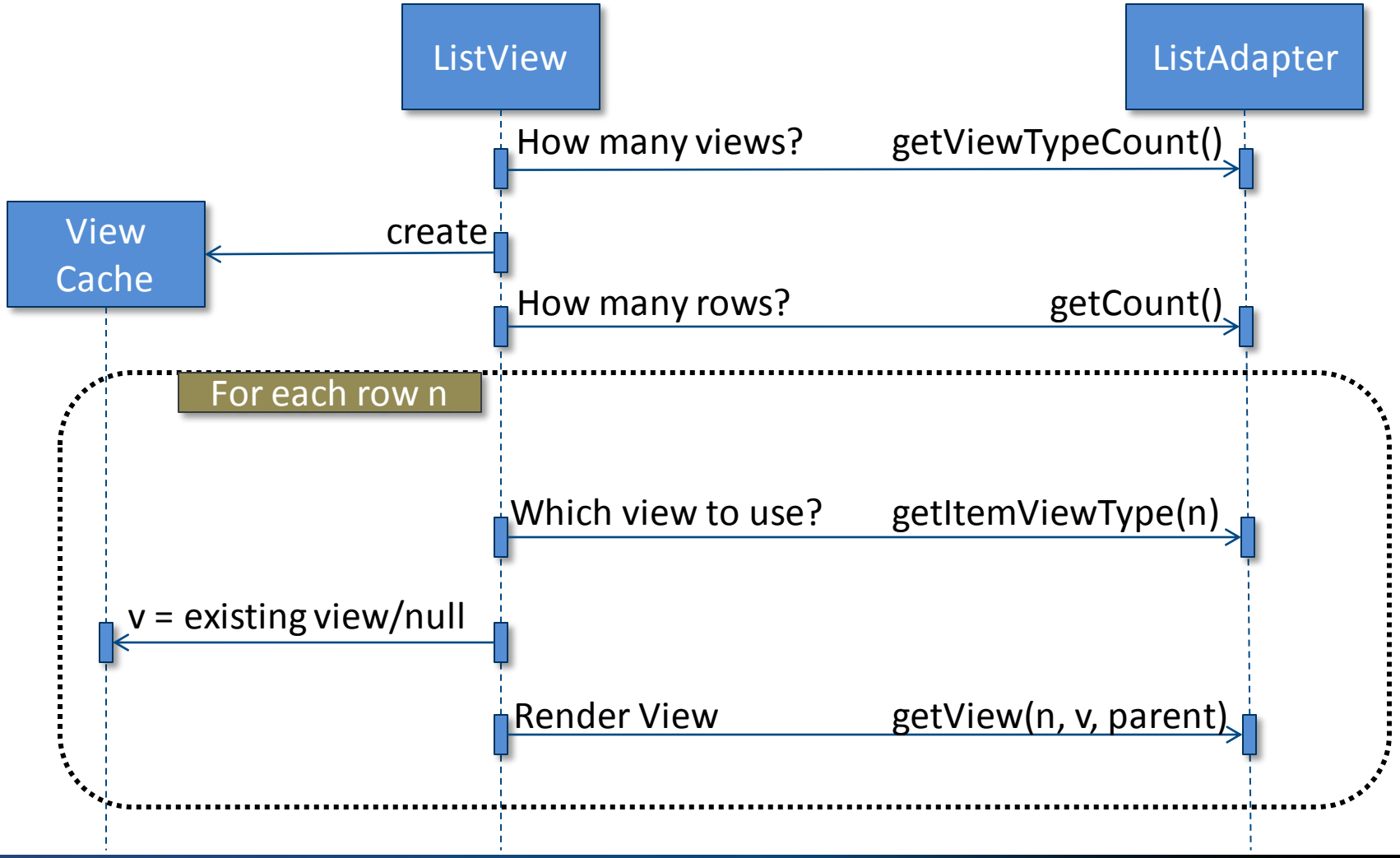
Row Rendering

- Rows can appear the same or different
- Row represented by View (or ViewGroup)
- ListView asks ListAdapter
  - How many different views are there?
  - What view should I use for row *n*?
- ListView caches Views and tries to reuse

Row Rendering – View Reuse

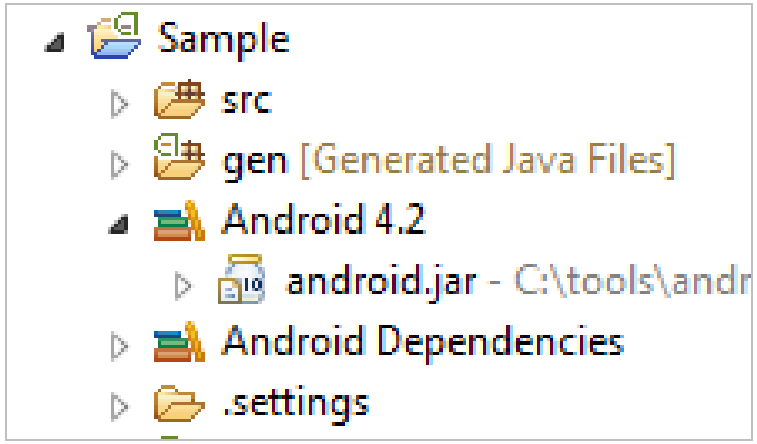


Row Rendering



Attaching Android Source

- Expand the Android library in your project
- Right-click on android.jar
- Choose “Properties”



- Under “Java Source Attachment”
  - Select “External Folder”
  - Find Android SDK/sources/Android-nn

Using ListView

```
ListView listView = (ListView) findViewById(R.id.listView1);

final String[] data = { "A", "B", "C"};

ArrayAdapter<String> arrayAdapter =
        new ArrayAdapter<String>(this,
                android.R.layout.simple_list_item_1, data);

listView.setAdapter(arrayAdapter);

listView.setOnItemClickListener(new OnItemClickListener() {
    @Override public void onItemClick(AdapterView<?> parent, View view,
                                      int position, long id) {
        Log.d("List", "Clicked: " + position);
    }
});
```